

Optimal Control and Closed-Loop Stability of Droplet Transport in a Microchannel

Rajneesh Anand¹ and Mayuresh V. Kothare^{1*}

¹Department of Chemical and Biomolecular Engineering, Lehigh University, Bethlehem, PA 18015, USA

Abstract

Understanding the efficient transport of fluid droplets in confined geometries has been a domain of interest for industrial applications in recent times. In the present work, we focus on designing control strategies that optimally steer droplet motion. Here, we apply optimal control framework to the droplet transport problem in a microchannel based on lubrication theory that minimizes viscous dissipation. Two complementary modeling routes are adopted: a reduced-order ordinary differential equation (ODE) model optimized via Pontryagin’s Maximum Principle, and a full nonlinear partial differential equation (PDE) model optimized using a Covariance Matrix Adaptation-Evolutionary Strategy. By parameterizing target displacement, droplet size, and capillary number, we uncover two distinct optimal transport regimes: a “translate–relax” strategy for short distances and a “compact–translate–relax” strategy for longer targets. In continuum mechanics, the competition between surface forces and cumulative viscous dissipation decides the optimal transport strategies. We further show that the displacement range over which the reduced-order controller transfers to the continuum model is governed by capillary stiffness. We finally address closed-loop stability by adapting a control Lyapunov function (CLF) framework to the reduced-order dynamics. We demonstrate that the terminal cost which penalizes deviation from the target, serves as a cost-compatible CLF on the physical domain, and prove that the CLF-compatible feedback exponentially stabilizes the target state.

I. INTRODUCTION

In recent years, transport and dynamics of fluid interfaces in single phase and multiphase have been areas of major research interest in many engineered systems, ranging from biological processes [1], microfluidics [2], and interfacial physics [3] to turbulent regime [4], [5] and confined bluff-body flows [6]. When such systems operate far from equilibrium, their dynamics become highly nonlinear and sensitive to spatiotemporal controls [1]. This motivates the use of optimal control as an interpretable framework to design spatiotemporal actuation strategies that balance transport efficiency, deformation, and cost. A broad class of navigation problems involving guiding aircraft or drones has been formulated to minimize transport time or energy [7]. Recently, a similar framework has been extended to micro-swimmers and active particles. Owing to their small size and driving force, these systems are influenced by the surrounding flow and strategically use it to reach target [8]. On the multicellular scale, patterned substrates and external fields have enabled controlled migration of monolayers which offers new possibilities for the control of active biological assemblies [9], [10].

Classical optimal transport theory, originating from Monge [11] proposed a highly nonlinear-nonconvex problem of moving mass from pile of earth to fill the holes of earth, later, Kantorovich relaxed the problem by adding local mass conservation and provided the direction to move mass or fluid distributions between given states while minimizing energetic cost over past centuries [12]. Later, Benamou-Brenier shapes the problem into a computational fluid dynamics and provided a different domain of interest where optimal control can be utilized for dynamical systems [13]. Still there was a lack of proper theoretical study which unifies geometrical aspects with complex dynamical system, Villani has huge contribution to shape the optimal control protocols in term of theoretical contributions [14]. However, these studies focus on isolated agents or multi-agents [15], [16] settings, and several other studies, such as, how to optimally guide extended, deformable materials that evolve through complex fluid domain [17], [18], thermal transient dynamics [19], and model predictive control in membrane microreactors [20]. In Ref. [17], [21], it has been studied for active drops (e.g., bacteria) which have an internal activity to drive the drops. A recent review [18] confirms that optimal control in soft and active matter has been predominantly applied to systems with internal driving mechanisms; active Brownian particles steered by light [22], [23], active nematics controlled via spatiotemporal activity fields [24], [25], and contractile gels modulated by optogenetic signals [26], [27]. All of which rely on the system’s intrinsic activity rather than an externally imposed pressure or flow field.

For a physical fluidic system where the control input is a pressure gradient or an externally imposed gradient rather than intrinsic activity, this class of problem remains unexplored. This gap leads to an unresolved question: how can we design and explore optimal strategies to efficiently transport fluids (droplets) under external fields in a microchannel? Here, we address this problem in the setting of translating a droplet through controlled spatiotemporal driving force. The droplet movement is governed by lubrication theory and capillary effects. By utilizing these constraints into the formulation and defining transport cost in terms of energetic dissipation and deformation penalties, we pose the question: how to optimally move droplet in a microchannel.

However, finding the optimal control for droplet movement is not enough, one must also ensure that the controller is stable, since a finite-horizon optimal strategy can fail to reach the target under perturbations. Stability of finite-horizon optimal control

*Corresponding Author: mvk2@lehigh.edu

is not automatically guaranteed; classical counterexamples show that optimality over a finite horizon can produce unstable closed-loop behavior [28]. The resolution, developed by Jadbabaie [29], employs a control Lyapunov function (CLF) as the terminal cost to ensure closed-loop stability without explicit terminal constraints. This framework has been widely adopted in nonlinear systems and model predictive control [19], [30], and extended to safety-critical settings [31]. In microfluidic systems, feedback control of droplets has been explored via PI control [32], AI-assisted methods [33], and model predictive control (MPC) of slug flow [34], but these approaches lack optimal control and formal stability analysis. Optimal control of active nematics [21], [24], [25] focuses on computing open-loop controls without stability analysis. Despite growing interest, CLF-based stability analysis for optimal microfluidic controllers has not been addressed.

To make this problem tractable and interpretable, we project the continuum droplet dynamics onto a lower order of modes, leading to a low-dimensional system of ordinary differential equations (ODEs) describing two states (position and shape) of the droplet. We have interestingly observed two distinct optimal transport regimes, corresponding to translate–relax (TR) and compact–translate–relax (CTR) strategies. Here, CTR is similar to gather–move–spread strategy for active drops [21]. These strategies arise from the competition between transport efficiency and transport cost. Full numerical simulations of the continuum model (partial differential equations (PDEs)) coupled with a gradient-free evolutionary optimization algorithm, confirming the qualitative predictions of the reduced-order description. We then provide a complete stability analysis for reduced-order model (ODE) by adapting the CLF framework [29]. Specifically, we establish: (i) the terminal cost is a valid CLF on the physical domain; and (ii) the CLF-compatible feedback yields exponential stability to the target. Our analysis exploits the driftless, control-affine structure of the droplet dynamics. Together, these results provide an interpretable framework for understanding optimal droplet transport and the trade-offs between efficiency, control efforts, and robustness.

II. MATHEMATICAL MODEL FOR OPTIMAL DROPLET TRANSPORT

A. Depth-integrated Continuity and Flux

We begin with incompressible flow $\nabla \cdot \mathbf{u} = 0$ (SI Appendix). In the microchannel geometry, the droplet is described by a height field $h(x, t)$ that measures the local thickness of the droplet phase. We describe a two-dimensional (2D) droplet dynamics moving axially (in the x -direction) on a thin film of oil on the solid surface (see Fig. 1), illustrated by lubrication theory [17], [21], [35], [36]. Here the droplet is separated from the channel walls by a continuous phase (oil), so that the lubricating film regularizes contact-line motion rather than eliminating it; the residual contact-line physics is retained in the PDE model through a precursor film and a disjoining pressure (SI Appendix). We neglect gravity by considering the drop size to be smaller than the capillary length. Integrating the continuity equation across the film thickness and applying the kinematic condition at the interface gives the depth-averaged continuity equation (Eq. 1) (SI Appendix),

$$\frac{\partial h}{\partial t} + \frac{\partial q}{\partial x} = 0 \quad (1)$$

$q(x, t) = \int_0^h u(x, z, t) dz$ is the horizontal volumetric flux per unit width. We specify conservation of volume

$$\int h(x, t) dx = 1 \quad (2)$$

In the lubrication limit (for low Reynolds number), the flow is governed by Stokes balance in the horizontal direction (as $H \ll W \ll L$; H , W , and L represent height, width, and length of the channel, respectively) $-\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial z^2} = 0$. The pressure consists of capillary curvature and control. We evaluate the velocity, $u(x, z, t) = \frac{z^2 - hz}{2\mu} \left(\frac{d}{dx} (\gamma h_{xx} + Gh) \right)$, where μ is the viscosity. Integrating it over the film thickness gives the flux (Eq. 3) (Here, $h_x = \partial_x h$, $h_{xx} = \partial_x^2 h$, & $h_{xxx} = \partial_x^3 h$)

$$q(x, t) = -\frac{h^3}{12\mu} (\gamma h_{xxx} + \partial_x (G(x, t)h)) \quad (3)$$

Here, $G(x, t)$ is the control. It is an imposed axial pressure gradient (equivalently, the driving force per unit volume), which may arise from any drive coupling to a property contrast between the droplet and the carrier phase including hydrostatic, centrifugal, magnetic, or dielectrophoretic. In the current model, Eq. 1 & Eq. 3 together with boundary and initial condition determines the droplet motion once the control is specified. For finite droplets, additional conditions near the contact line, such as partial slip, prewetting films, or contact angles need to be included [37]. In this formulation, surface tension, γ sets the curvature of the interface, while the external pressure gradient (control) acts as a spatiotemporal driving term that modifies the effective flux. For convenience, we adopted a simplified form of control input. We describe the control using two unsteady parameters a mean $G_0(t)$ and a spatial variation $\Delta G(t)$. This creates a low-dimensional control space that is easy to interpret and suitable for optimization. We assume the droplet occupies a finite interval on the x -axis, with its left and right edges at $x = a(t)$ and $x = a(t) + R(t)$, respectively. The midpoint of the droplet is therefore $x_c(t) = a(t) + \frac{R(t)}{2}$ (Fig. S1). We impose linear variation of the control (Eq. 4) to introduce a spatial dependence

$$G(x, t) = G_0(t) + \Delta G(t) \left(\frac{x - x_c(t)}{R(t)} \right) \quad (4)$$

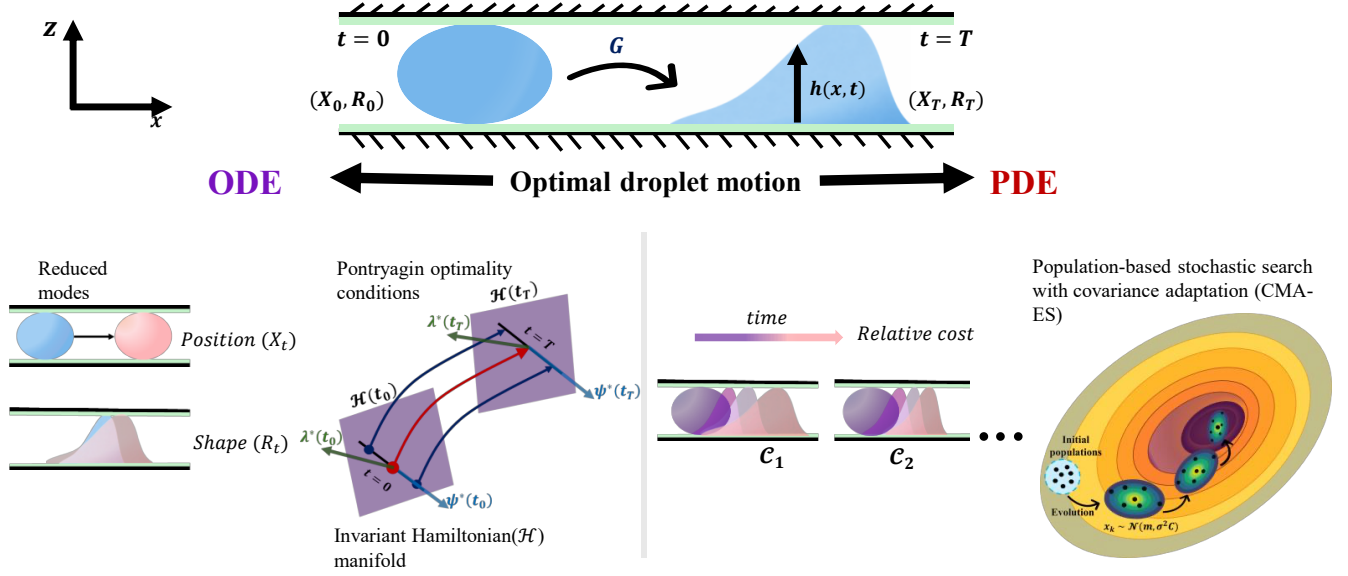


Fig. 1: A schematic illustration of optimal motion of a droplet from its initial position and size (X_0, R_0) to a target state (X_T, R_T) while minimizing a cost, such as viscous dissipation in a microchannel. The control (G) shapes $h(x, t)$. We utilize two parallel approaches: (i) a reduced-order ODE model derived from position and shape modes and solved using Pontryagin's Maximum Principle (PMP), and (ii) a full PDE formulation optimized with a gradient-free evolutionary strategy (CMA-ES) using adaptive Gaussian sampling $(\mathcal{N}(m, \sigma^2 C))$, where m = mean, σ^2 = variance, C = covariance matrix).

However, the droplet motion is passive, and its evolution arises from a competition between externally imposed viscous stresses and restoring forces due to surface tension. The relative strength of these two effects can be measured using a capillary number (Eq. 5). Small values $Ca_G \ll 1$ correspond to surface-tension-dominated regime, the droplet remains nearly spherical; $Ca_G \gg 1$ shows strong deformation driven by the imposed pressure gradient. For pressure-driven motion, we define an effective capillary number (**SI Appendix**)

$$Ca_G = \frac{1}{T} \int_0^T \frac{|\Delta G| R(t)^2}{\gamma} dt \quad (5)$$

B. Optimal Transport Formulation

The objective of optimal droplet transport is to identify a transient control input that drives the droplet from its initial state to a target state while minimizing a chosen objective (cost) function. In our formulation, the total cost $\mathcal{J} = \mathcal{W} + \mathcal{T}$: $\mathcal{W}$ measures the mechanical effort exerted on the system during the motion (Eq. 6), and a terminal cost $\mathcal{T}$, which penalizes any disagreement between the observed and desired state (Eq. 7). Under the lubrication approximation, the dominant source of dissipation arises from viscous shear, scales as $\mu(\partial_z u)^2$. The accumulated mechanical work can be expressed as an integral over shear-induced dissipation.

$$\mathcal{W} = \int_0^T dt \int \left(\frac{h^3}{12\mu} (\partial_x (Gh + \gamma \partial_{xx} h))^2 \right) dx \quad (6)$$

$$\mathcal{T} = \mathcal{A} \left(\frac{X(T) - X_T}{X_T} \right)^2 + \mathcal{B} \left(\frac{R(T) - R_T}{R_T} \right)^2 \quad (7)$$

The energetic cost associated with the droplet translation is always non-negative, because viscous dissipation cannot generate energy. Moreover, we consider a fixed transport duration T , although other formulations such as minimal-time control are possible. To encode the objective of droplet movement to a target distance and achieve a desired final shape, we add a terminal penalty (Eq. 7). We show that $\mathcal{T}$ satisfies the conditions of a control Lyapunov function (CLF) for the reduced-order dynamics, with an explicit exponential convergence rate in **section VI**. X_T and R_T denote the target center of mass (COM) position and drop size (variance) at the end of the control horizon. $\mathcal{A}$ and $\mathcal{B}$ are terminal cost weighting coefficients. The droplet position is defined through its COM, $X(t) = \frac{\int x h(x, t) dx}{\int h(x, t) dx}$, using Eq. 2, we have $X(t) = \int x h(x, t) dx$ and $R(t)$ measures its effective size. These quantities are evaluated at $t = T$ to assess how well control achieves the desired task.

C. Optimal Control Framework

With the known governing equations and formulated cost functional, we now outline optimal control strategy. A central challenge in this problem is that the dynamics are nonlinear, the cost functional is non-convex, and the control enters multiplicatively through the flux, coupling the state and control in a way that precludes standard linear quadratic methods. We